EXPERIMENT E7 · VARIANT C · MEASUREMENT REVIEW → REVISION B (FRAME REV. B3)

Will the deformation measurement work?

27 Oct 2026 · read as the person who will run the tests · numbers from the axisymmetric heat model (sim_thaw.py), indicative

Verdict. The rig will run, and temperature, TDR and Pt100 will be measured without trouble. Deformation is the delicate part: in the unsaturated sand (Sr = 0.5, decided 9 Nov) the expected heave is about as large as the thermal drift of the rig itself, so E7 measures mainly the thaw settlement under load. Revision B removes most of the rig-side problems.

Findings and what revision B changes

TOPIC	PROBLEM IN REV. A	CHANGE IN REV. B	EFFECT	STATUS
Signal size	Sr = 0.5: ice adds 1.7 % volume, pore air is 19 % → in a closed sample the expansion fits into the pores	none on the rig; Sr 0.5 chosen (see below)	expected in-situ heave ≈ 0 (tens of µm at most); upper bound 1.7 mm	✓ decided
Rig drift	LVDT on the Al beam with a steel rod; camera referenced to the PETG rim	Invar 36 post from the base frame carries the LVDT and a camera target; Invar guide rod	per 20 K: LVDT −24…+48 → −59…0 µm, left after blank-run correction 12 → ≤ 6 µm; camera 0…100 → −49…0 µm, left 8 → ≤ 3 µm	✓ fixed
Freezing direction	cell on a 25 mm Al plate freezes from below and the sides too; last unfrozen spot 35 mm above the bottom	cell printed in one piece ⌀240 × 200: +40 mm low-infill wall and base (λ ≈ 0.06), 3 G10 pads in base pockets; frame of T-slot profiles, housing under the top beam, beam gap 140 → rig 559 mm high	freezes from the top: last unfrozen 13 mm above the bottom; 95 % frozen in 15 h (was 10.5 h)	✓ fixed
Laser	45° laser: collar shadows a 25 mm strip, 41° DOE gives only 7–8 lines	laser at 66° beside the top beam, 200 mm from the surface (DOE253: 3.1° between lines, 31° fan – Picotronic quote)	no collar shadow; ≈ 9 lines on the sand at 11.9 mm, each ≈ 111 mm long; line shift 0.44 Δh	✓ fixed
Camera	CSI ribbon cannot pass the lid seal; separate heater near the sample	RPi 5 inside the camera box (its ≈ 5 W keep the window clear); the RPi takes photos on its NTP-synced clock; 6 mm lens	Ethernet + 5 V through the seal only; time stamps within < 1 s of the CR850	✓ fixed
Bearing	LM12UU ball bushing: grease stiffens at −10 °C, ice at thaw → stick-slip, load not constant	igus drylin RJ4JP-01-12, dry polymer, same size	load change from friction < 0.02 kPa	✓ fixed
Moisture	open surface sublimates while frozen; frost and dew at the start of thaw look like heave	witness dish with the same sand weighed each cycle (the cell stays in place, weighed only at the start and end of a series); SHT45 air RH on the RPi	loss and frost become visible in the data	✓ added
Freezer air	on/off control swings ±2–3 K; air sensor only optional	80 mm fan; Campbell 109 air probe required	boundary condition known; smaller drift noise	✓ added

✓ Decided 9 Nov: Sr = 0.5

E7 measures the **thaw settlement under load** in unsaturated sand.

- **Expect tens of µm** per cycle; rely on the blank-run correction (LVDT ≤ 6 µm, camera ≤ 3 µm left).
- **Frost heave** would need Sr ≥ 0.85 or water supplied from below – not part of E7.

Before the first real cycle

1. Blank run: Al dummy instead of sand, one full cycle; fit drift against cell temperature.
2. Calibrate the camera with shims 0.1 / 0.5 / 1.0 mm under a flat plate.
3. Measure the freezer: rig 559 mm vs ≈ 650 mm inside.
4. Confirm LVDT operation at −10 °C with RDP / Elkome.
5. Program: 24 h at −10 °C, 24 h at +10 °C (thaw takes ≈ 15 h).

Cost and resolution

≈ 1,460–1,990 €
rev. B excl. VAT (rev. A 1,365–1,870 €); frame from T-slot profiles cut to length and printed parts, no machining

≤ 6 µm
LVDT drift left after the blank-run correction (model), LVDT resolution ≈ 1 µm

≈ 10 µm
camera height map per laser line after averaging ≈ 1000 px along the line

Details: drawings E7-V01…V05 rev. B, purchase list, calculations §15 (vypocty.md), 3D model with drift chart: E7 sample thawing.

Model assumptions: Sr 0.5, θ 0.19, n 0.38, λ sand 1.3 / 2.0 W/mK, h = 7 W/m²K in still air; frame parts as lumped elements. Treat all numbers as estimates to be checked by the blank run.